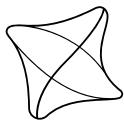


UNIVERSITY OF ZILINA
FACULTY OF ELECTRICAL ENGINEERING AND INFORMATION
TECHNOLOGY
Department of Multimedia and Information and Communication Technologies

0123456789

TITLE OF WORK



ŽILINSKÁ UNIVERZITA V ŽILINE
Fakulta elektrotechniky
a informačných technológií

Katedra multimédií a
informačno-komunikačných technológií

Title of work

Bachelor's thesis

Field of study: Informatics

Study program: Information and Communication Technologies

Thesis supervisor: prof. Ing. Juraj Machaj, PhD.

Žilina, 2026

Titl. Meno Priezvisko

Zadanie práce

Zadanie práce je na osobitnom liste od vedúceho katedry a tento list ním musí byť nahradený!

ANOTAČNÝ ZÁZNAM - BACHELOR'S THESIS

Meno a priezvisko: Titl. Meno Priezvisko

Akademický rok 2025/2026

Názov práce:

Title of work

Počet strán: 20

Počet obrázkov: 20

Počet tabuliek: 20

Počet grafov: 20

Počet príloh: 20

Použitá literatúra: 20

Anotácia v slovenskom jazyku:

Anotácia je krátka charakteristika, stručný popis informačného zdroja nasledujúci bibliografickú citáciu. Anotácia môže obsahovať sumár kľúčových bodov zdroja, opis druhu zdroja a jeho ohodnotenie. Na rozdiel od abstraktu, čo je krátky objektívny popis zdroja, anotácia je kritické alebo subjektívne hodnotenie diela zvyčajne zahrnuté v bibliografii alebo citácii.

Anotácia v anglickom jazyku:

An annotation is a brief description of the information source followed by a bibliographic citation. An annotation may include a summary of the key points of the source, a description of the type of source, and an evaluation of the source. Unlike an abstract, which is a brief objective description of a source, an annotation is a critical or subjective evaluation of a work usually included in a bibliography or citation.

Kľúčové slová: Kľúčové slová

Thesis supervisor: prof. Ing. Juraj Machaj, PhD.

Recenzent diplomovej práce: doc. Ing. Meno Priezvisko, PhD.

Dátum odovzdania práce: 27. apríl 2026

PodĎakovanie

Uvádza sa tu podĎakovanie, vrátane oznamu o riešení práce v rámci projektu (ak existuje).

Táto práca bola riešená v rámci nasledujúcich výskumných úloh a projektov:

- Projekt č. 1
- Projekt č. 2

Abstrakt

Abstrakt obsahuje informáciu o cieľoch práce, jej stručnom obsahu a v závere abstraktu sa charakterizuje splnenie cieľa, výsledky a význam celej práce. Súčasťou abstraktu je tri až päť kľúčových slov. Abstrakt sa píše súvisle ako jeden odsek a jeho rozsah je spravidla 100 až 500 slov.

Kľúčové slová: Kľúčové slová

Abstract

The abstract contains information about the aims of the work, its brief content and the conclusion of the abstract characterizes the fulfillment of the aim, the results and the significance of the whole work. The abstract includes three to five key words. The abstract is written coherently as a single paragraph and is usually 100 to 500 words in length.

Keywords: Keywords

Obsah

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Zoznam skratiek

Skratka	Anglický význam	Slovenský význam
SPM	Self-Phase Modulation	Vlastná fázová modulácia
Skratka 2	Anglický význam	Slovenský význam

Skratka	Anglický význam	Slovenský význam
SPM	Self-Phase Modulation	Vlastná fázová modulácia
Skratka 2	Anglický význam	Slovenský význam

Zoznam symbolov

[illegible]

Introduction

V úvode autor stručne a výstižne charakterizuje stav poznania alebo praxe v oblasti, ktorá je predmetom záverečnej práce a oboznamuje čitateľa s významom, cieľmi a zámermi záverečnej práce. Autor v úvode zdôrazňuje, prečo je záverečná práca dôležitá a prečo sa rozhodol spracovať danú tému.

- Introduction - motivation of the work
- modulations - theory
 - division of modulations (analog and digital - differences)
 - FM modulation (analog)
 - digital modulations (QAM, PSK ...)
- AMC - state of the art
 - why it is needed
 - how it is done
- experimental setup
 - -> dataset overview
 - -> overview of models
- Proposed solution
- results

1 The State Of Art

First, I state that the focus of this thesis is Automatic Modulation Classification (AMC), an essential component of RF wireless communication receivers. The core challenge of this thesis is to compare fixed or classical machine learning architectures with new or advanced deep learning architectures. There are several reasons why AMC is a significant problem in wireless communication. Firstly, we know that the RF spectrum is expensive, as we are unable to increase the bandwidth. Even if that were possible, it would also increase the noise of the channel, which would significantly decrease the channel's effectiveness. Additionally, we have problems in different fields like spectrum management, where, depending on channel quality, we may need to set lower order modulation and higher coding rate for better BER (bit error rates) or higher order modulation with less coding rate when we have better channel quality. Another important field is interference detection, where jamming signals may have a different modulation type than those authorized transmission signals, which are very common attacks in the Internet of Things (IoT).

The timeline of the approaches used so far starts from likelihood-based extraction, after which they perform featured-based methods which improved this process because they didnt require high computational complexity, however, featured-based methods do have other tradeoffs like the limited learning capacity, these because ML classification methods like Support Vector Machines or Decision Trees contain fixed architectures, meaning that the parameters they take into account for classification are fixed input data. This is a big change in deep learning models like Convolutional Neural Networks, where the different values used to set and configure the architecture of the model change with biases and weights, this when the model uses backpropagation.

- **Classical Machine Learning (ML)** for this thesis covers two stages:

1. **Extraction:** Here we will work with cumulants or high-order statistic values like skewness or kurtosis.
2. **Classification:** Here, we expect to input this information to the ML supervised learning type model. In our case, we plan to use *Support Vector Machines (SVMs)* and *Decision Trees (DT)*.

- **Advanced Machine Learning (ML):**

1. Here we carry out **deep learning** with distinct possibilites, in this thesis we will work with neural networks (NN), more specifically, with convolutional neural networks (CNNs), where both extraction and classification happen in the model, but we may have different filters (layers) where we evaluate different characteristics from the raw I/Q sequence input signal data. Meaning that the extraction of the data mainly happens through backpropagation, and the classification happens each time the data pass through the layers of the CNN.

This thesis focuses on the problem of **Automatic Modulation Classification (AMC)** of radio-frequency (RF) signals in the presence of channel noise. AMC plays a critical role in modern wireless communication systems, including cognitive radio, spectrum monitoring, and interference identification.

The objective of this thesis is to design and evaluate a machine-learning-based system that classifies modulation types from complex baseband I/Q samples under varying Signal-to-Noise Ratio (SNR) conditions. Special emphasis is placed on comparing the robustness of **traditional feature-based machine learning approaches** with **deep learning-based methods**.

1. **The output** from the models implemented will be the modulation type prediction, so they must yield which type of modulation is used by the signal going through the model.
2. **The input** for the *Traditional feature-based machine learning models*:
 - Feature-based extracted values from the signal, amplitude, phase offset, frequency, timing.
 - Statistical houses: according to the article, distinct values from statistics, like distributions, cumulants.
 - High order cumulants: mean, variance, skewness, and kurtosis.
3. **The input** for the *Advanced Deep learning Machine Learning models*:
 - I/Q In Phase and Quadrature Samples since they are the principal type of data composing the dataset from RadioML2018 for 1D CNN.
 - Amplitude and Phase - real and imaginary values for 1D CNN.
 - We could implement a 2CNN fusion with Modulation Constellation Diagrams, which is mainly not recommended due to its complexity for interpretation, thus our model would become a black box model, where we aren't able to explain what is truly happening inside it.

There are several constraints for designing a AMC deep learning model, for instance:

- One of the first constraints for our models can be having only one channel, and ignoring multipath channels in which we are passing the signals, which limits the real application of AMC for real-life environments. So, for this thesis, we assume only one channel configuration for easier model configuration.

In classical machine learning models, susceptibility to noise and indifference often occur because models learn from noise patterns, which may lead to overfitting. Here, the usage of optimization with the loss function's numeric values is often applied to improve the model.

Another essential limit that we can highlight is the lack of automatic learning, since the models are based on pre-disposed data, which means that once input, we are not able to change it; these inputs are often higher-order cumulants, statistical measurements from a distribution that we are typically set up by experts, and at the end, they are just fixed numeric values.

In our specific case (Advanced Machine Learning Models), we also face some tradeoffs like high computational complexity, since we need good storage and computer performance, so that we can run these models. Additionally, fusing different types of data to train the same model, for instance, 2D modulation constellation diagrams as input in the same model with 1D convolution to input our sequential I/Q samples or amplitude and phase complex values,

1.1 Data overview

According to different sources, DeepSig Dataset: RadioML 2018.01A is a good experimental database, which is described by the DeepSig team website as "a dataset which includes both synthetic simulated channel effects of 24 digital and analog modulation types, which has been validated. This dataset was used in our paper Over-the-air deep learning-based radio signal classification which was published in 2017 in IEEE Journal of Selected Topics in Signal Processing, which provides additional details and description of the dataset." from which we can deduce, is a trustworthy dataset, although they recommend researches to create their own, I needed to take into account that creating a dataset with over 2 million of samples and stratify samples correctly along with SNR values would take much more time, which can be used for the model training, since here falls the purpose of my work, intending to demonstrate how deep and supervised learning can develop a way better approach than classical machine learning methods.

The dataset is balanced across modulation classes because each class has an identical number of samples (4,096 per SNR per class), meaning we are able to trust the classical 0.5 approach for the model classification threshold, thus we will be able to analyze the different metrics and imply knowledge from them.

There will be different metrics for the models; among the best, we will evaluate a confusion matrix and, from it, calculate precision, recall, F1 score, and accuracy, which are among the best metrics for evaluating deep learning methods.

The evaluation is performed using the publicly available **RadioML dataset**, which contains digitally modulated signals affected by realistic channel impairments.

1.2 Work methodology and research methods

Part „Work methodology and research methods“ usually contains:

1. Characteristics of the object of investigation:

- (a) There are different characteristics taken into account for the methodology and research methods.

2. Work Procedures:

- (a) Working with DeepSig Dataset: RadioML 2018.01A, which were created with the purpose of research and experimentation.

3. Method Of Obtaining Data And Its Sources:

- (a)

4. Methods Used For Evaluation And Interpretation Of Results:

(a)

5. Statistical Methods:

(a)

1.3 Results of the work and discussion

Výsledky práce a diskusia sú najvýznamnejšími časťami záverečnej práce. Výsledky, ktorými sú najmä vlastné postoje alebo vlastné riešenie vecných problémov, ku ktorým autor dospel, autor logicky usporiada a pri popisovaní ich dostatočne zhodnotí. Zároveň autor komentuje všetky skutočnosti a poznatky v konfrontácii s výsledkami iných autorov. Ak je to vhodné, výsledky práce a diskusia môžu tvoriť aj jednu samostatnú časť a spoločne tvoria spravidla 30 až 40 záverečnej práce.

2 Ukázka formátovania práce

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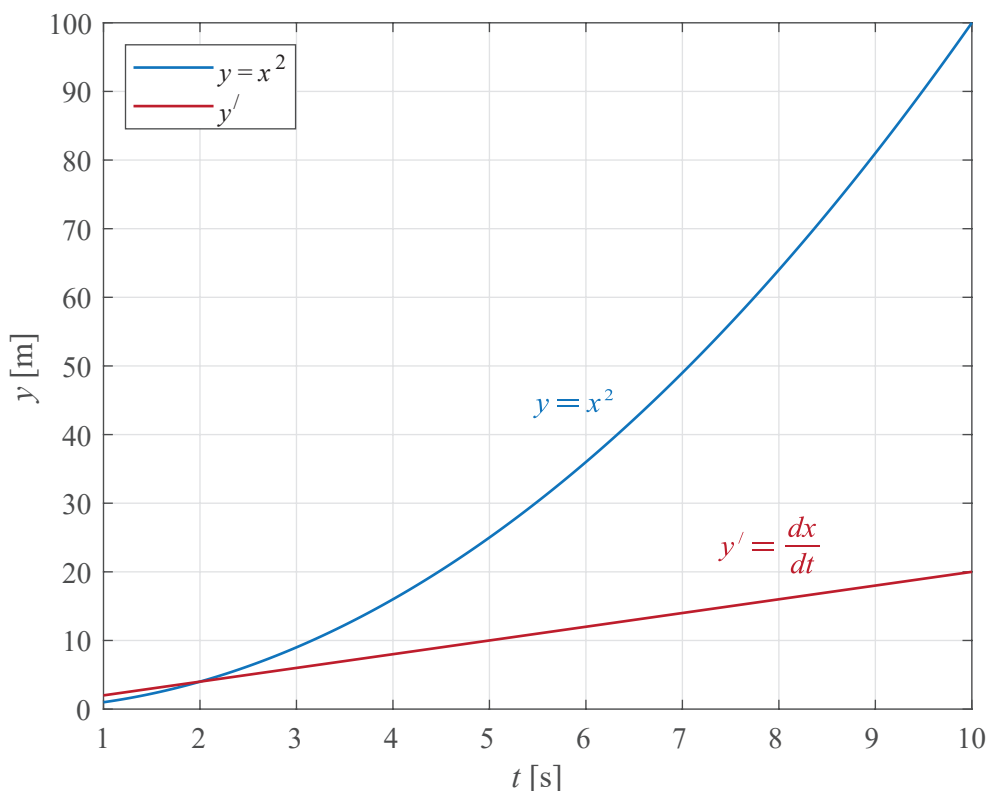
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Obr. 2.1: Ukážka formátovania obrázka.

Ukážka formátovania matematického vzťahu

$$y = \frac{d^2y}{dx^2}, \quad (2.1)$$

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Tab. 2.1: Ukážka formátovania tabuľky.

Delenie	DWDM	CWDM
$\Delta\lambda_1$	100 GHz	20 nm
$\Delta\lambda_2$	50 GHz	20 nm

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Záver

V závere autor v stručnosti zhrnie dosiahnuté výsledky vo vzťahu k stanoveným cieľom. Ak je záverečná práca vypracovaná v cudzom jazyku, obsahuje resumé v slovenskom jazyku v rozsahu spravidla 10 % rozsahu záverečnej práce.

Literatúra

- [1] Michal Frniak, Patrik Kamencay, Miroslav Markovic, Jozef Dubovan, and Milan Dado. Comparison of Vehicle Categorisation by Convolutional Neural Networks using MATLAB. In *ELEKTRO 2020 PROC.*, pages 1–4, Taormina, 2020. IEEE. ISBN 978-1-7281-7542-3.

Čestné vyhlásenie

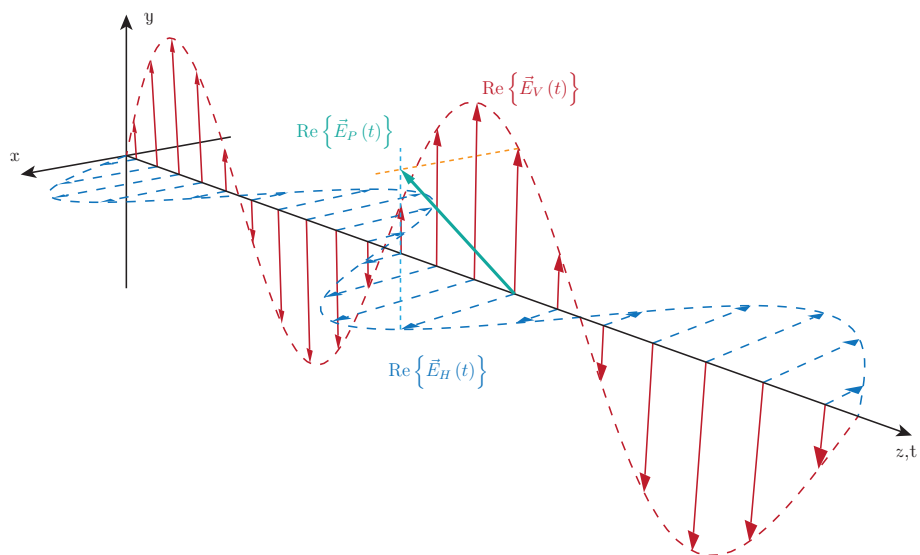
Čestne vyhlasujem, že som zadanú diplomovú [bakalársku] prácu vypracoval(a) samostatne, pod odborným vedením vedúceho diplomovej [bakalárskej] práce a používal(a) som len uvedenú literatúru.

Žilina, 13. mája 2026

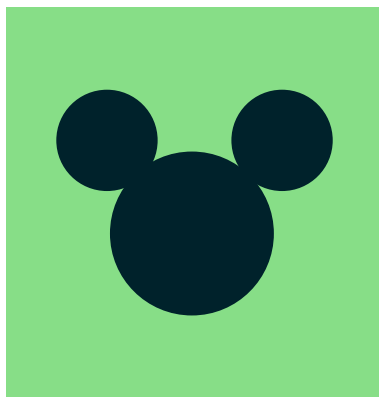
Podpis autora

Prílohová časť

Príloha A



Obr. A.1: Popis obrázka A.



Obr. A.2: Popis obrázka B.

Príloha B

Sem je potrebné vypísať prílohy alebo vkladať obrázky, hyperlinky alebo zdrojové kódy.